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Preconception and early-pregnancy exposure to ambient air pollution and trajectories of depressive symptoms during pregnancy

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Abstract

Background Prenatal depression shows diverse patterns in severity and progression. While air pollution has been linked to depression risk, its impact on the course of prenatal depression remains understudied. The study objectives were to (1) identify distinct trajectories of depressive symptoms across pregnancy, and (2) to evaluate whether preconception and early-pregnancy air pollution exposure impacts membership in trajectory groups.

Methods This study followed 542 predominantly low-income Hispanic/Latina participants in the MADRES cohort. We assessed depressive symptoms using the Center for Epidemiologic Studies-Depression (CES-D) scale at 1st, 2nd, and 3rd trimester visits. Daily residential concentrations of PM_{2.5}, PM₁₀, NO₂, and O₃ were estimated using inverse-distance squared spatial interpolation from monitoring data. We used latent growth mixture modeling (LGMM) to identify trajectory classes and multinomial logistic regression to estimate adjusted relative risk ratios (RRRs) of membership in each class relative to the lowest group with averaged air pollution levels over preconception and early pregnancy.

Results We identified four distinct trajectories: moderate-to-high symptoms (7.4%), consistently-moderate symptoms (14.6%), moderate-to-low symptoms (34.7%), and consistently-low symptoms (43.4%). Higher proportions of Black mothers and those not cohabitating were found in the two highest symptom trajectories. Participants with an interquartile range higher early-pregnancy NO₂ (10.28 ppb) had a 3.82-fold higher relative risk (95% CI: 1.50–9.73) of following the moderate-to-high versus consistently-low-symptom trajectory. Similar elevated risks were observed for PM_{2.5} (RRR = 1.61; 95%CI: 1.08–2.40) and PM₁₀ (RRR = 1.84; 95% CI: 1.01–3.35).

Conclusion Early-pregnancy exposure to ambient air pollution is associated with increased risk of moderate-to-high depressive symptoms across pregnancy.

Keywords Depression trajectory, Prenatal depression, Air pollution, Early pregnancy, Latent growth mixture modeling

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Introduction

Perinatal depression is a significant public health concern and is marked by symptoms such as persistent sadness, restlessness and impaired concentration during pregnancy and in the first months after childbirth [1–3]. Global estimates suggest that 10–20% of women experience perinatal depression [4, 5], with approximately 17% of women experiencing symptoms during pregnancy. Among those with prenatal symptoms, approximately 39% of women subsequently develop postpartum depression [6]. Certain populations face disproportionate risk, with U.S. Hispanic/Latina women experiencing additional vulnerabilities including limited access to mental health services, limited Spanish-speaking mental health care providers, and acculturative stress among immigrant populations [7–10]. Given this substantial burden and disparities in risk, identifying modifiable factors for prenatal depression represents an urgent priority, particularly among these vulnerable subgroups.

Emerging epidemiological and mechanistic evidence implicates ambient air pollution exposure during preconception and early pregnancy as a potential risk factor for prenatal depression. A retrospective cohort of more than 200,000 pregnant women from 19 hospitals across the US showed that preconception and first trimester exposure to ambient pollutants, including nitrogen dioxide (NO₂) and particulate matter, were associated with increased odds of maternal depression during pregnancy [11]. A study based in the Shanghai Maternal-Child Pairs Cohort (Shanghai MCPC) of over 3700 pregnant women further demonstrated that first-trimester particulate matter with aerodynamic diameter less than 2.5 µm (PM_{2.5}) exposure was associated with a 25% increased risk of late-pregnancy depression [12]. Mechanistically, studies suggest that prenatal exposure to air pollution may contribute to depressive symptoms through activation of the hypothalamic–pituitary–adrenal (HPA) axis, as evidenced by elevated cortisol levels—a pathway linked to stress response and mood dysregulation [13]. Emerging evidence suggests that the HPA axis undergoes hormonal regulations throughout the menstrual period and early pregnancy [14], thus, environmental exposures, including air pollution, during these periods may increase a women's susceptibility to developing prenatal depression.

While existing studies provide important evidence linking air pollution to prenatal depression, primarily examined at a single time point, they overlooked the dynamic nature of depressive symptoms throughout pregnancy. Emerging evidence suggests that the course of maternal depressive symptoms throughout pregnancy exhibit considerable heterogeneity, characterized by diverse patterns of onset, progression, duration, and severity [15–17], which potentially reflect different underlying etiologies and environmental vulnerabilities [18]. Environmental

exposures like air pollution may differentially affect specific trajectory patterns depending on timing and critical exposure windows, revealing more nuanced relationships than cross-sectional analyses are able to discern. Recent advances in trajectory analysis, such as latent class trajectory modeling (LTCM) methods, allow investigation of patterns of depressive symptoms throughout the perinatal period and thus identify factors that may predict these patterns [19].

Despite accumulating evidence linking ambient air pollution to mental health outcomes including perinatal depression, no study to date has explicitly focused on the trajectories of depressive symptoms across pregnancy in relation to preconception and early-pregnancy pollution exposure (NO₂, PM_{2.5}, particulate matter with aerodynamic diameter less than 10 µm (PM₁₀), and ozone (O₃)). We hypothesized that particularly early pregnancy represents a critical window for environmental influences on prenatal depression trajectories, as this period is characterized by dramatic neuroendocrine and hormonal changes, including rapidly rising human chorionic gonadotropin (hCG) between 3–4 and 9–10 weeks' gestation [20, 21] and the luteal-placental progesterone shift at 6–9 gestational weeks [21]. Notably, existing research has largely been cross-sectional or has focused on postpartum outcomes, without modeling symptom trajectories across pregnancy or considering early-gestational exposure windows explicitly. In addition, limited research has focused on environmental health disparity populations or those with high burden of exposure to air pollution and disparities in prenatal mental health.

To address these gaps, we conducted a prospective study comprised of primarily low-income Hispanic/Latina women in urban Los Angeles, CA. This study has two primary objectives: (I) to identify latent trajectories of prenatal depressive symptoms using an advanced LTCM approach called latent growth mixture modeling (LGMM) that enables detection of heterogeneous symptom patterns across pregnancy; and (II) to evaluate associations between ambient air pollution exposure during preconception and early-pregnancy and these identified prenatal depression trajectories.

Methods

Study population and study design

The Maternal and Developmental Risks from Environmental and Social Stressors (MADRES) study is a prospective pregnancy cohort of women were recruited before 30 weeks of gestation at three community health clinics. A description of the cohort protocol can be found in a previous publication [22]. Eligibility criteria included English or Spanish fluency and over 18 years old. Exclusion criteria included multiple gestation, incarceration, and physical, mental, or cognitive disability that

prevented informed consent. Written informed consent was obtained at the time of recruitment in English or Spanish. The Institutional Review Board at the University of Southern California approved all aspects of this study.

The participants included in this analysis were recruited from November 2015 through February 2023. The consort diagram of the analytic study population is shown in Supplemental Fig S1. Among 1065 consented mothers, 865 eligible participants had one or more (up to three total) prenatal depression assessments during pregnancy visits (during 1st, 2nd, 3rd trimesters). Since a minimum of two time points was required for trajectory modeling, we excluded 323 mothers who had only one depression assessment during pregnancy. Therefore, the final analytical sample for trajectory analysis comprised 542 mothers (Part I). When analyzing the association with air pollution, we further excluded samples based on data availability for ambient air pollution exposure levels ($N=109$) and covariates ($N=5$). We also excluded 5 participants who had first depressive symptoms assessment measured before 6 weeks of gestation (the window of air pollution exposure in early pregnancy), leading to 423 mothers included in the association analyses (Part II).

Assessment of ambient air pollution and temperature

We constructed and geocoded daily residential time-lines for each participant based on residential addresses and occupancy dates obtained through residential history forms from 2 years prior to the third trimester visit, along with prospective address verification data collected at every follow-up visit to track residential mobility [23]. We used inverse-distance-squared weighted spatial interpolation to estimate daily levels of 24-h NO_2 , 24-h $\text{PM}_{2.5}$, 24-h PM_{10} , and 8-h maximum ground-level O_3 at each participant's residential locations, using data from the United States Environmental Protection Agency Air Quality System monitors [23]. The study area, Southern California, has the densest air pollution monitoring network in the U.S., with an average of four monitoring stations located within 8–14 km for exposure estimation. The daily temperature was estimated by calculating the average of the maximum and minimum temperatures recorded each day, which were obtained from a high-resolution gridded surface (4km $\times$ 4 km) meteorological dataset [24].

Average levels of ambient air pollutants and ambient temperature were calculated across different exposure windows for assessment. Specifically, we defined preconception (up to 12 weeks prior to conception) and early pregnancy (6 weeks after conception) as the exposure windows of interest. The 12-week preconception window was selected based on its biological relevance for menstrual cycles and ovulation processes that may influence pregnancy [25, 26]. The 6-week early pregnancy window

was chosen to ensure that the exposure assessment period preceded the earliest prenatal outcome score measurement.

Measurement of depressive symptoms during pregnancy

Depressive symptoms were assessed based on the Center for Epidemiological Studies-Depression (CES-D) Scale [27] at 1st, 2nd and 3rd trimester visits during pregnancy administered by trained bilingual staff. The CES-D scale is a 20-item validated tool for screening depressive symptoms in the prior week. Responses to each item were scored on a 0 to 3 Likert scale. A CES-D score (range: 0–60) of 16 or higher indicated clinically-relevant symptoms, as recommended by the authors of the CES-D scale [27]. We used continuous CES-D scores as the outcome in the trajectory modeling (Part I), with higher scores indicating a greater severity of depressive symptomatology. For the air pollution models (Part II), we used the trajectory groups of CES-D scores identified from Part II as the outcome.

Covariates

We included potential covariates selected *a priori* using a Directed Acyclic Graph (DAG) [28] and based on prior work [29] (Supplemental Fig S2) for part II analyses. These covariates were the confounders associated with both preconception and early-pregnancy exposure to ambient air pollution and different trajectory groups of prenatal depressive symptoms, and were further determined by minimal sufficient adjustment set. Interviewer-administered questionnaires (English/Spanish) collected demographic, socioeconomic status, housing characteristics, and behavioral information at study entry and during the follow-up period. Previous depression history was defined by any depression diagnosis or history of antidepressant medication recorded in prenatal medical records (EMR). Time of conception was estimated as date of birth minus gestational age at birth (in days). Gestational age at birth was calculated and standardized using a hierarchy of methods consistent with established clinical guidelines [30]: first trimester crown-rump length ultrasound was the preferred method; when unavailable, second trimester fetal biparietal diameter ultrasound was used; for the remaining participants, gestational age was established from the physician's best clinical estimate recorded in the EMR. The covariates adjusted for included maternal age, recruitment site, race and ethnicity, maternal education at cohort entry, annual household income at cohort entry, previous depression history, household air conditioning use during pregnancy, season of conception, and temperature levels averaged over the same period of the exposure to ambient air pollution. We did not adjust for gestational age, pregnancy complications (e.g., Hypertensive disorders of pregnancy (HDP),

gestational diabetes mellitus (GDM)) since they might mediate the association between ambient air pollution and trajectory of depressive symptoms during pregnancy.

Part I: latent growth mixture modeling (LGMM)

Descriptive analyses were conducted to examine the characteristics of the study population. We also explored the patterns of the CES-D scores for each individual and average course of CES-D scores among all samples during pregnancy using spaghetti plots overlaid with LOW-ESS smoothing curves to reveal potential non-linear patterns to be assessed in the trajectory models (Supplemental Fig S3).

To identify distinct trajectories of maternal depressive symptoms during pregnancy, we employed LGMM, a LTCM approach. Unlike traditional mixed-effect models that assume population homogeneity and describe a single mean trajectory with individual variability, LTCM approaches assume a heterogeneous population comprised of unknown subgroups (latent classes), each characterized by a unique mean trajectory profile [31–33]. LTCM aims to minimize within-group variance while maximizing between-group differences, ensuring greater similarity within groups than between groups [34]. We specifically chose LGMM over other LTCM approaches (e.g., group-based trajectory models) because it allows for variability between individuals within subgroups, effectively combining standard mixed-effects models with LTCM [34, 35].

CES-D scores on the original continuous scale were modeled as either linear or quadratic functions of assessment time. Specifically, assessment time points were labelled following the approach from previous literature [19] with delivery as the reference time point. CES-D scores measured in the third trimester were assigned 0 years, and equal time interval were assigned to the three timepoints. We performed a grid search across each combination of the following model specifications [34, 36] over different number of classes (2–5): trajectory shape (CES-D ~ year or CES-D ~ year² + year), random effects structures (no random effects; random intercept only; random slope + random intercept; random quadratic effect + slope + intercept), and variance–covariance matrix specifications across classes (equal or unconstrained). We allowed differences in residual variances over time without autocorrelation processes for computational convergence reasons. Since there is always a risk of a mixture-model solution at a local rather than a global maximum in mixture modeling, all models are run multiple times (800 iterations) using random initial values [34, 35]. Across these runs, the solution with the highest log likelihood is identified.

The optimal number of classes within each model specification was selected based on the Lo-Mendell-Rubin

(LMR)-likelihood ratio test (LRT), where a p -value < 0.05 indicated improved model fit when comparing a k -class model versus a $(k-1)$ -class model [37]. Among all model specifications with their respective optimal number of classes, the final model was selected based on the lowest Bayesian Information Criterion (BIC) [36, 37], in addition to a comprehensive framework that has been proposed previously to identify a good-fitting LGMM [34, 36]: (1) average posterior class membership probabilities (AvePP) should be more than 0.7 for each class to indicate goodness of fit; (2) sufficient number of individuals in each subgroup (more than 5%) should be achieved; (3) relative entropy (E_k) for the best fitted model should exceed 0.6 to suggest good discrimination ability; (4) biological plausibility of the optimal classification was assessed by inspecting predicted trajectory subgroup. We also conducted a sensitivity analysis to evaluate if the classification was sensitive to the unbalanced CES-D datapoints by repeating the whole procedure using the samples restricting to mothers who had all 3 measurements of CES-D scores ($N = 392$).

Part II: multinomial logistic regression models

We fit multinomial logistic regression models to estimate the relative risk ratio (RRR) of membership in each trajectory group relative to the trajectory group with the lowest symptoms with respect to an interquartile range (IQR) increase in the average levels of ambient air pollution exposure across different windows during preconception and early pregnancy periods. Specifically, we assessed exposure during the following windows: (1) early pregnancy only (averaged over 0–6 gestational weeks), (2) preconception only (–12–0 weeks and –6–0 weeks, respectively, with the latter chosen for comparability with the 6-week gestational period window), and (3) combined periods (0–6 gestational weeks + –12–0 preconception weeks, and 0–6 gestational weeks + –6–0 preconception weeks, respectively; the latter combination ensured that the 12-week preconception window did not override the 6-week early pregnancy exposure averaging). Covariates adjusted in the models are listed in the Covariates section. Multicollinearity among covariates was assessed using the variance inflation factor (VIF), with a threshold of VIF < 10 considered acceptable [38].

We conducted sensitivity analyses to examine the robustness of our findings: (1) We extended the exposure window up to 12 gestational weeks, and modeled average exposure levels for 0–7, 0–8, 0–9, all the way through 0–12 gestational weeks, respectively, in relation to different trajectory groups using multinomial logistic regression models. To maintain temporal relationships, we restricted analytical samples to participants whose first CES-D measurement occurred after the corresponding exposure window cutoff, resulting in a slightly decreased

sample size for each additional gestational week included. (2) We conducted sensitivity analyses including only the exposure window from conception to the gestational week before the first CES-D measurement during pregnancy. (3) We performed the analyses using exposure averaged over the entire pregnancy period to evaluate the

Table 1 Participants characteristics for the analytic samples for Part I depression trajectories^a, Part II air pollution models^b

Mean (SD)/N (%)	Analytic sample for depression trajectories (N=542)	Analytic sample for air pollution models (N=423)
Maternal Age at Consent, years	29.0 (5.9)	28.9 (6.0)
Race/Ethnicity		
Hispanic	436 (80.4%)	336 (79.4%)
White, non-Hispanic	27 (5.0%)	25 (5.9%)
Black, non-Hispanic	53 (9.8%)	45 (10.6%)
Others	21 (3.9%)	17 (4.0%)
Missing	5 (0.9%)	0
Education Attainment		
High school or below	290 (53.5%)	229 (54.1%)
Some college or technical school	151 (27.9%)	121 (28.6%)
4 years of college or above	96 (17.7%)	73 (17.3%)
Missing	5 (0.9%)	0
Annual Household Income		
< \$30,000	257 (47.4%)	198 (46.8%)
\$30,000 to \$49,999	63 (11.6%)	52 (12.3%)
\$50,000 or more	62 (11.4%)	51 (12.1%)
Not known	160 (29.5%)	122 (28.8%)
Parity		
0	189 (34.9%)	143 (33.8%)
1 or more	334 (61.6%)	266 (62.9%)
Missing	19 (3.5%)	14 (3.3%)
Pre-pregnancy Depression Diagnosis or Antidepressant Medication		
Not recorded	452 (83.4%)	375 (88.7%)
Yes	53 (9.8%)	48 (11.3%)
Missing	37 (6.8%)	0
Ever Use of Household AC During Pregnancy		
No	183 (33.8%)	151 (35.7%)
Yes	359 (66.2%)	272 (64.3%)
Missing	0	0
Season at Conception		
Cold	274 (50.6%)	209 (49.4%)
Warm	253 (46.7%)	214 (50.6%)
Missing	15 (2.8%)	0
Pre-pregnancy BMI, kg/m ²	29.0 (7.0)	28.7 (6.4)

^aPart I population were participants who had two or more measurements of CES-D scores during pregnancy

^bPart II population were participants who had two or more measurements of CES-D scores during pregnancy with the first assessment after gestational week 6 (this was to ensure that the exposure assessment period preceded the earliest prenatal CES-D score measurement), who also had estimates of exposure levels of preconception and early-pregnancy ambient air pollution, and who did not have missing information of key covariates

Abbreviations: SD Standard deviation, AC Air conditioning, BMI Body mass index

possible concurrent effects of air pollution exposure on depressive symptom trajectories throughout pregnancy. (4) We additionally adjusted for severity of nausea and vomiting in pregnancy (NVP) during early pregnancy as serious NVP has been associated with increased depressive symptoms [39], assessed using the 24-h Pregnancy-Unique Quantification of Emesis and Nausea (PUQE-24) [40] scale at the first trimester visit. The PUQE-24 is a validated, reproducible scoring system that evaluates NVP severity based on three symptoms over a 24-h period: duration of nausea, frequency of vomiting, and frequency of retching [40]. Scores range from 3 to 15, with higher scores indicating more severe symptoms [40]. The total PUQE score was used to classify NVP severity as mild (3–6 points, $N=268$), moderate (7–12 points, $N=115$), or severe (≥ 13 points, $N=8$) [40, 41]. (5) We additionally adjusted for parity to examine whether the results were sensitive to this adjustment (nulliparous women: $N=143$; women with one previous birth: $N=131$; women with two or more previous births: $N=135$). (6) We performed the multi-pollutant model where we adjusted for averaged levels of concurrent O_3 when we assessed the effect estimates for the exposure to NO_2 , $PM_{2.5}$, and PM_{10} . Similarly, when assessing the effect estimate for the exposure to O_3 , we adjusted for concurrent exposure to NO_2 . (7) We re-performed analyses excluding participants who delivered after March 1, 2020 ($N=373$), to assess the potential impact of the COVID-19 pandemic on our findings. (8) We re-performed analyses excluding participants who had pre-pregnancy depression diagnosis or antidepressant use in the medical records ($N=375$), to evaluate the influence of the pre-pregnancy depression on our results.

Data were managed in SAS and statistical analyses were performed in R. The *Mplus* software was used to perform LGMM. We chose *Mplus* for its modeling flexibility, allowing differences in residual variances over time, correlated residuals across time points, and different variance–covariance matrices of random effects per class [42]. Also, *Mplus* uniquely provides statistical tests comparing fit improvement between neighboring class models [42].

Results

Descriptive analyses

Participant characteristics for the Part I depression trajectory analysis ($N=542$) and Part II association analysis with air pollution ($N=423$) are presented in Table 1. For Part I, mean maternal age was 29.0 years ($SD=5.9$) at study entry. Most participants were Hispanic/Latina (80.4%), with a high school education level or below (53.5%) and annual household income $< \$30,000$ (47.4%). About 9.8% mothers had a previous depression diagnosis or antidepressant medication use. As for parity,

34.9% of pregnancies were first-born. The distribution of these characteristics in the Part I study population was very similar to that in the Part II study population (Table 1). Compared to all 865 mothers who had at least one CES-D measurement during pregnancy, we did not observe substantial deviation and statistically significant difference in the pattern of characteristics for the Part I and Part II study populations (Table 1 and Supplemental Table S1).

We found a non-linear trend in average CES-D scores over pregnancy, with the highest scores (median=10, IQR=11) and highest proportion (26%) of mothers who scored at or above the threshold of 16 for clinically-relevant depression during the 1st trimester visit (12.4 ± 4.4 gestational weeks). This was followed by a leveling off during the second trimester visit (CES-D scores median=8, IQR=9, measured at 20.8 ± 3.1 gestational weeks) and third trimester (median=9, IQR=9, measured at 31.5 ± 1.8 gestational weeks) (Supplemental Fig S3), with the proportions of clinically-relevant depression remaining at 19% and 17%, respectively. The distributions of average exposure levels to NO_2 , O_3 , $\text{PM}_{2.5}$, and PM_{10} were similar across different exposure windows, including preconception, early pregnancy, combined windows over both preconception and early pregnancy, and the entire pregnancy period, except that exposure during the entire pregnancy period showed less variation due to averaging over a longer period of time (Supplemental Table S2). Correlations across exposure windows were moderate to strong for most pollutants, with the exception of $\text{PM}_{2.5}$, which showed only slight correlation between the preconception and early gestation periods (Supplemental Figure S4). Across pollutants, $\text{PM}_{2.5}$ and O_3 were slightly negatively correlated, and NO_2 and O_3 were strongly negatively correlated, while all remaining

inter-pollutant correlations were positive (Supplemental Figure S5). These patterns were consistent across both the preconception and early-pregnancy exposure windows (Supplemental Figure S5). Also, the four pollutants generally followed consistent seasonal variation patterns across years where air pollution estimates were available (2015–2020), with the exception of a notable fluctuation in $\text{PM}_{2.5}$ and PM_{10} in early 2018 (Supplemental Fig S6).

Part I: Identification of different trajectory classes of depressive symptoms during pregnancy

By evaluating all combinations of model specifications described in the methods (Supplemental Table S3 and Table S4), we observed that the 4-group LGMM with quadratic time function, random intercept and slopes, and unconstrained variance–covariance matrices over 4 classes, yielded the best model fit satisfying all evaluation criteria (Supplemental Table S4).

According to the optimal selected model, we identified four distinct trajectories of depressive symptoms during pregnancy, with mean trajectories for each class shown in Fig. 1. Most MADRES pregnant women ($N=235$, 43.4%) had consistently low depressive symptoms during pregnancy, followed by those with moderate-to-low depressive symptoms ($N=188$, 34.7%). For these two largest trajectory classes, throughout pregnancy, mean CES-D scores remained below 16, the suggested cut-off for clinically relevant depressive symptoms [27]. A smaller proportion of participants had consistently-moderate depressive symptoms ($N=79$, 14.6%), with mean CES-D scores hovering around 16 throughout pregnancy (slightly above or below 16). Forty pregnant women (7.4%) were identified as having moderate-to-high depressive symptoms, characterized by an inverted U-shaped mean trajectory of CES-D scores with the highest average depressive symptom levels observed at mid-pregnancy well above 16, and mean CES-D scores hovering around 16 during the 1st and 3rd trimesters.

Trajectory class membership characteristics are presented in Table 2. We observed higher proportions of Black mothers and those with pre-pregnancy depression diagnosis or antidepressant medication use in the two higher symptom trajectory classes (moderate-to-high and consistently-moderate depressive symptoms) compared to the relatively low symptom trajectory classes (moderate-to-low and consistently-low symptoms). Additionally, the highest proportion of mothers not cohabitating with a partner (40%) was found in the moderate-to-high symptoms class compared to the other three trajectories. No other notable differential patterns across the four classes were observed.

In sensitivity analyses (Supplemental Table 5 and Table S6), we also observed that a 4-class LGMM with quadratic time function, random intercept and slopes, and

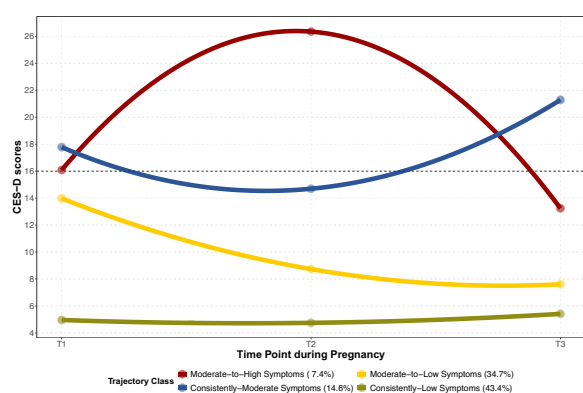


Fig. 1 Mean trajectory of depressive symptoms over pregnancy within each of the four different classes identified by the LGMM of best fit ($N=542$). The dashed line represents the suggested cutoff for clinically-relevant depressive symptoms (CES-D ≥ 16). Abbreviations: Latent Growth Mixture Modeling (LGMM); Center for Epidemiological Studies-Depression (CES-D) Scale

Table 2 Demographic characteristics and maternal depression risk factors by trajectory class for mothers in the four identified classes

Mean (SD)/N (%)	Moderate-to-high (N=40)	Consistently-moderate (N=79)	Moderate-to-low (N=188)	Consistently-low (N=235)
Maternal Age at Consent, years	28.5 (6.2)	28.5 (5.0)	28.5 (6.1)	29.6 (5.9)
Race/Ethnicity				
Hispanic	32 (80.0%)	56 (70.9%)	150 (79.8%)	198 (84.3%)
White, non-Hispanic	1 (2.5%)	4 (5.1%)	11 (5.9%)	11 (4.7%)
Black, non-Hispanic	6 (15.0%)	15 (19.0%)	19 (10.1%)	13 (5.5%)
Others	1 (2.5%)	4 (5.1%)	5 (2.7%)	11 (4.7%)
Missing	0 (0%)	0 (0%)	3 (1.6%)	2 (0.9%)
Education Attainment				
High school or below	19 (47.5%)	39 (49.4%)	101 (53.7%)	131 (55.7%)
Some college or technical school	13 (32.5%)	32 (40.5%)	54 (28.7%)	52 (22.1%)
4 years of college or above	8 (20.0%)	8 (10.1%)	30 (16.0%)	50 (21.3%)
Missing	0 (0%)	0 (0%)	3 (1.6%)	2 (0.9%)
Annual Household Income				
< \$30,000	23 (57.5%)	38 (48.1%)	101 (53.7%)	95 (40.4%)
\$30,000 to \$49,999	5 (12.5%)	10 (12.7%)	20 (10.6%)	28 (11.9%)
\$50,000 or more	5 (12.5%)	6 (7.6%)	19 (10.1%)	32 (13.6%)
Not known	7 (17.5%)	25 (31.6%)	48 (25.5%)	80 (34.0%)
Parity				
0	15 (37.5%)	26 (32.9%)	68 (36.2%)	80 (34.0%)
1 or more	25 (62.5%)	50 (63.3%)	114 (60.6%)	145 (61.7%)
Missing	0 (0%)	3 (3.8%)	6 (3.2%)	10 (4.3%)
Pre-pregnancy Depression Diagnosis or Antidepressant Medication				
Not recorded	30 (75.0%)	57 (72.2%)	160 (85.1%)	205 (87.2%)
Yes	8 (20.0%)	19 (24.1%)	15 (8.0%)	11 (4.7%)
Missing	2 (5.0%)	3 (3.8%)	13 (6.9%)	19 (8.1%)
Cohabitation status				
Cohabiting	22 (55.0%)	51 (64.6%)	130 (69.1%)	181 (77.0%)
Not cohabiting	16 (40.0%)	23 (29.1%)	48 (25.5%)	43 (18.3%)
Declined/Missing	2 (5.0%)	5 (6.3%)	10 (5.3%)	11 (4.7%)
Gestational age at birth, weeks	39.1 (1.6)	38.9 (1.8)	38.9 (1.8)	39.0 (2.1)
Pre-pregnancy BMI, kg/m ²	29.7 (12.7)	29.7 (7.3)	28.6 (6.3)	28.8 (6.0)

Abbreviations: SD Standard deviation, BMI Body mass index

unconstrained variance–covariance matrices across classes yielded the best model fit, satisfying all evaluation criteria (Supplemental Table S6). Notably, this optimal model retained identical specifications and identified the same number of trajectory classes as the primary analysis model. Also, the four trajectory patterns were similar to those found in primary analyses (Supplemental Fig S7), and membership assignments between the sensitivity and primary LGMM models showed substantial overlap (Supplemental Table S7).

Part II: The association between preconception and early-pregnancy exposures to ambient air pollution and membership of trajectory class of depressive symptoms during pregnancy

We observed significantly increased relative risks of moderate-to-high depressive symptoms compared to consistently-low depressive symptoms in relation to

increased early pregnancy (gestational weeks 0–6) exposure to NO₂, PM_{2.5}, and PM₁₀ (Fig. 2). Specifically, mothers with an IQR higher early-pregnancy NO₂ (10.28 ppb) exposure had a 3.82-fold higher relative risk (95% CI: 1.50, 9.73) of being in the moderate-to-high depressive symptoms trajectory class compared to being in the consistently-low depressive symptoms trajectory class. Similarly, mothers with an IQR higher early-pregnancy exposures to PM_{2.5} (3.00 µg/m³) and PM₁₀ (8.79 µg/m³) had 1.61-fold (95% CI: 1.08, 2.40) and 1.84-fold (95% CI: 1.01, 3.35) higher relative risk of being in the moderate-to-high depressive symptoms trajectory class compared to being in the consistently-low depressive symptoms trajectory class, respectively (Fig. 2). Both the magnitude and statistical significance of associations between moderate-to-high versus consistently-low depressive symptoms in relation to NO₂, PM_{2.5}, and PM₁₀ were attenuated when extending exposure windows to the preconception

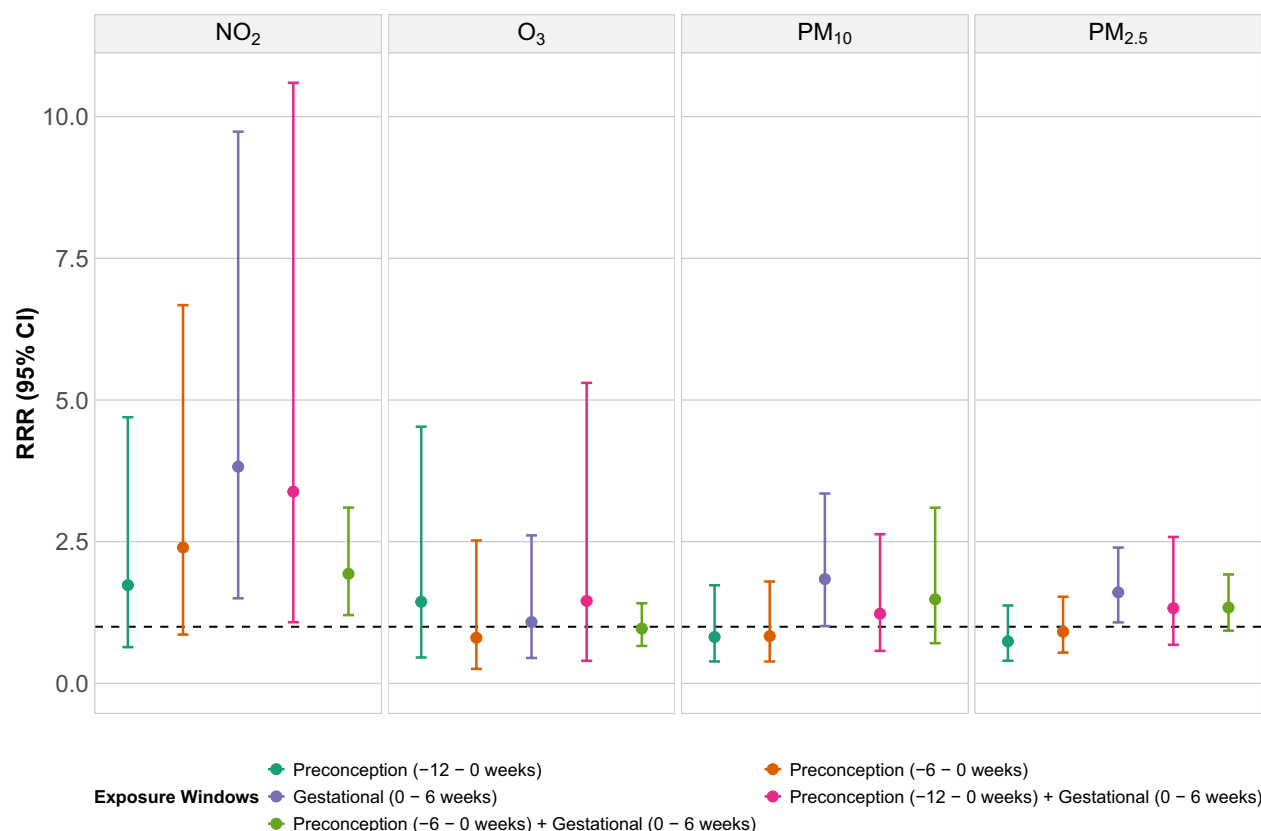


Fig. 2 Relative risk ratio (RRR) of membership in moderate-to-high depressive symptoms trajectory group relative to the consistently-low depression trajectory class per IQR increase in the averaged exposure levels to ambient air pollution over different exposure windows from 12 weeks preconception to 6 weeks gestation ($N=423$). These multinomial logistics regression models were adjusted for maternal age, recruitment site, race/ethnicity, education level, annual household income level, depression diagnosis history on medical record, air conditioning use during pregnancy, season of conception, and averaged levels of concurrent temperature. Abbreviations: relative risk ratio (RRR); interquartile range (IQR); nitrogen dioxide (NO_2); particulate matter with aerodynamic diameter less than $2.5 \mu\text{m}$ ($\text{PM}_{2.5}$) and less than $10 \mu\text{m}$ (PM_{10}); ozone (O_3)

period (12 or 6 weeks prior to conception) (Fig. 2). No suggestive association patterns were found for O_3 (Fig. 2). Largely null associations were observed for consistently-moderate and moderate-to-low depressive symptom trajectory classes compared to the consistently-low depressive symptom trajectory class across all exposure windows during preconception and early pregnancy for all four pollutants assessed (Supplemental Fig S8).

Sensitivity analyses confirmed the robustness of early pregnancy as a critical exposure window, revealing similar patterns of increased relative risks of membership in the moderate-to-high depressive symptoms trajectory class compared to the consistently-low depressive symptoms trajectory class for NO_2 , $\text{PM}_{2.5}$, and PM_{10} when extending early pregnancy exposure windows to 7, 8, 9, through 12 gestational weeks after conception (Supplemental Fig S9). Similar suggestive increases in relative risks persisted when we included exposure windows from conception to the first measurement timepoint (Supplemental Fig S9). We also observed potential concurrent effects of NO_2 , $\text{PM}_{2.5}$, and PM_{10} exposure throughout pregnancy on moderate-to-high depressive symptoms,

although these associations were not statistically significant (Supplemental Fig S9). Also, we did not observe substantial deviation of the patterns of the associations from the primary results after adjusting for severity of NVP (Supplemental Fig S10) and parity (Supplemental Fig S11). Additionally, the patterns of the associations from the primary results (single-pollutant model) persisted when performing multi-pollutant models (Supplemental Fig S12). Lastly, we did not observe the major change in patterns of the associations after excluding mothers who delivered after March 1 2020 (Supplemental Fig S13) and after excluding mothers who had pre-pregnancy depression diagnoses or antidepressant use in the medical records (Supplemental Fig S14).

Discussion

In this predominantly low-income Hispanic/Latina population, we identified four distinct trajectories of maternal depression during pregnancy, with most mothers exhibiting consistently low symptom trajectories throughout pregnancy. Among the highest symptom trajectory class of moderate-high-symptoms across pregnancy, we

observed a higher proportion of Black mothers suggesting there may be important differences in risk of sustained depression across pregnancy across demographic groups. We also observed higher proportions of mothers who were not living with a partner during pregnancy in the highest symptom trajectory class, which suggests that higher emotional social support from a partner during the pregnancy period may be an important protective factor. We also found novel evidence that early pregnancy exposure to higher levels of ambient NO_2 , $\text{PM}_{2.5}$, and PM_{10} was associated with increased risk of moderate-to-high depressive symptoms throughout pregnancy.

To our knowledge, this is the first study to estimate the effects of preconception and early-pregnancy ambient air pollution exposure on depressive symptom trajectories throughout pregnancy. Limited research has assessed associations between exposure to preconception or early pregnancy ambient air pollution exposure and prenatal depression incidence; however, our results are somewhat consistent with results from a large medical records-based retrospective cohort analysis across multiple US hospitals, finding that early pregnancy (1st trimester exposure) exposures to ambient NO_2 and PM_{10} were associated with 25% and 12% increased odds of maternal depression diagnoses per IQR increase of exposure, respectively [11]. The same study showed limited evidence for preconception NO_2 and PM_{10} exposures with 15% and 9% increased odds of maternal depression diagnoses per IQR increase in 3 month average preconception exposure which we did not find evidence for in our study. A cohort study in China further demonstrated that first-trimester $\text{PM}_{2.5}$ exposure was associated with a 25% increased risk of late-pregnancy depression [12]. Our findings extend these findings by incorporating longitudinal measures of depressive symptoms across pregnancy and finding that early-pregnancy air pollution exposure increases the risk of experiencing moderate-to-high depressive symptom patterns throughout the entire prenatal period, rather than at a single time point or relying on a diagnosis of depression in the medical record which is likely underdiagnosed and underreported in health disparity populations. While our findings did not provide evidence that preconception is a critical window for prenatal depressive symptom trajectories, future studies are needed to further investigate the preconception period as a potential important window in relation to prenatal maternal mental health.

We found novel evidence that early pregnancy exposure to higher levels of ambient air pollutants (NO_2 , $\text{PM}_{2.5}$, and PM_{10}) was associated with increased risk of moderate-to-high depressive symptoms across pregnancy. Notably, we observed pollutant-specific patterns in critical exposure windows. For NO_2 , the magnitude of association was strongest for exposure during gestational

weeks 0–6, with effects gradually attenuating as the exposure window extended from 0–7 through 0–12 gestational weeks. In contrast, we did not observe this attenuation pattern for $\text{PM}_{2.5}$ and PM_{10} . The effect estimates for $\text{PM}_{2.5}$ were largely consistent across 0–6 through 0–12-week exposure windows, while PM_{10} showed stronger associations for the entire first trimester rather than the first 6 weeks alone, though statistical significance was limited by reduced sample sizes in some analyses. The underlying mechanisms explaining these different sensitive windows remain unclear and warrant further investigation with large sample sizes throughout early pregnancy.

These pollutant-specific exposure window patterns may reflect distinct mechanistic pathways through which different air pollutants influence maternal mental health. Mechanistic research has implicated the role of air pollution in inducing neuroinflammation, oxidative stress, and cerebrovascular damage [43]. These processes have been linked to the development of depression due to disruptions in neurotransmitter and hormonal regulation [44]. The HPA axis, which plays a key role in neuroendocrine stress responses, can be activated by inhaled air pollution [45]. Emerging evidence has indicated the involvement of HPA axis activity in the pathophysiology of depression [46]. Particulate matter including $\text{PM}_{2.5}$ and PM_{10} can be inhaled into the lungs, leading to systemic inflammation, which can further trigger HPA axis activation [47, 48]. In contrast, NO_2 plays a pivotal role in starting a signaling cascade, subsequently leading to oxidative stress, and potentially contributes to depression and anxiety pathophysiology [49, 50]. These mechanistic differences may partially explain why NO_2 showed a clearer and narrower critical exposure window compared to particulate matter in our findings.

Our findings suggest that early pregnancy represents a more critical exposure window than the preconception period for air pollution effects on prenatal depression trajectories. This pattern may reflect the unique physiological changes occurring during early pregnancy, characterized by drastic neuroendocrine and hormonal adaptations due to the fetus-placental unit [21]. Early pregnancy is marked by dramatic hormonal fluctuations, including rapidly rising hCG levels between 3–4 and 9–10 weeks' gestation [20], peaking at approximately 10 weeks [21], and the critical luteal-placental progesterone shift occurring at 6–9 weeks when progesterone production transitions from the corpus luteum to the placenta [21]. Additionally, placental gonadotropin-releasing hormone (GnRH), a key hormone of HPA axis, peaks at approximately 8 weeks' gestation before declining [20, 21, 51]. We speculate that the pregnant women may be particularly vulnerable during this period of rapid hormonal change, as air pollution-induced dysregulation may

compound these ongoing neuroendocrine adaptations, potentially disrupting the delicate balance required for normal mood regulation. Another potential mechanistic pathway involves air pollution-induced thyroid disruption during early pregnancy, with evidence suggesting that gestational weeks 5–8 represent a particularly sensitive window for PM_{2.5} effects on maternal thyroid function [52]. Thyroid dysregulation during this critical period may subsequently disrupt the HPA axis, as thyroid hormones play essential roles in neuroendocrine regulation. Both thyroid hormone excess and deficiency can alter HPA axis activity, with excess activating and deficiency suppressing the axis [53–55], and such HPA axis dysregulation is closely linked to prenatal depression risk [46]. Similarly, thyroid metabolism has also been implicated in maternal depression in the context of effects of other environmental exposures, including polybrominated diphenyl ethers (PBDEs) [56], and poly- and perfluoroalkyl substances (PFAS) [56, 57]. However, the precise mechanisms linking early pregnancy air pollution exposure to prenatal depression trajectories remain unclear, warranting further research on potential mediators, including thyroid regulation, and the interplay between multiple neuroendocrine processes during this critical developmental window.

The study also found that a higher proportion of Black women and women not living with a partner were represented in the highest prenatal depressive symptom trajectory class, consistent with well-documented social disparities in maternal mental health. Black mothers experience a higher prevalence of perinatal mood disorders, including postpartum depression and anxiety, compared to U.S. national estimates [58]. Fractured healthcare systems, limited access to resources, insufficient universal screening, and a shortage of culturally congruent providers together limit Black mothers' access to mental health services [59]. Black mothers are also more likely to experience single parenthood and workplace discrimination, each of which independently elevates vulnerability to perinatal depression [60]. Although Hispanic/Latina ethnicity was not identified as a risk factor for the highest symptom trajectory class in our predominantly Hispanic/Latina MADRES cohort, Hispanic/Latina women broadly face elevated risk for maternal mental health disorders, driven by immigration stress, cultural stigma, language barriers, and limited access to mental healthcare services [7–10]. These structural challenges are similar to those experienced by Black mothers. Therefore, these findings underscore the role of racial and ethnic disparities in the burden of prenatal depression, and highlight the need for further investigation into the structural, cultural, and other factors that drive these inequities.

This study has several strengths. First, we employed latent growth mixture modeling to identify distinct phenotypes of prenatal depressive symptom trajectories, offering several advantages over traditional approaches. Unlike single time-point assessments or cross-sectional designs, trajectory modeling characterizes whether depressive symptoms are persistent, increasing, decreasing, or curvilinear across pregnancy, capturing clinically meaningful heterogeneity that a single CES-D measurement cannot detect. Importantly, distinct trajectory phenotypes may reflect different underlying etiologies. This distinction has important implications in the present study: the two highest trajectory classes had nearly overlapping symptom scores at the first trimester and would likely have been treated as a single elevated group in prior cross-sectional studies, however, they exhibited different patterns of symptom progression, and different associations with early-pregnancy air pollution exposure, suggesting they represent etiologically distinct subgroups. The trajectory approach also avoids reliance on depression diagnoses from medical records, which are likely underdiagnosed and underreported in low-income health disparity populations, and instead characterizes the course of depressive symptom across the entire prenatal period. Second, our comprehensive assessment of exposure windows, including preconception periods combined with early pregnancy, allowed for identification of critical exposure windows and evaluation of cumulative effects. Third, we captured detailed residential history data from a year prior to conception to birth, enabling more accurate exposure assessment than studies relying solely on birth addresses.

Several limitations should be considered when interpreting our findings. First, selection bias may exist as participants included in the final analytical sample differed from the broader MADRES population (including those without two or more depression evaluations during pregnancy). However, we expect this bias to be minimal given the comparison of demographic characteristics between those included in the present study and the broader MADRES cohort showed similar distributions. Second, participants were recruited from the urban Los Angeles area, which may limit generalizability to other geographic regions with different pollution sources, meteorological conditions, or population characteristics. Third, not all participants had CES-D measurements at all three time points during pregnancy, with some having only two assessments. This missing data resulted in reduced sample sizes for trajectory modeling and potential bias. However, sensitivity analyses restricted to participants with complete CES-D data across all three trimesters demonstrated very similar trajectory patterns, confirming the robustness of our classification approach to missing data. Fourth, the CES-D scale we used to measure the

depressive symptoms is a screening instrument rather than a diagnostic tool; therefore, we interpreted high scores as reflecting elevated depressive symptomatology rather than clinically diagnosed depression. Fifth, given the growing evidence that ambient air pollution is associated with depression in non-pregnant populations, residual confounding by pre-pregnancy depressive symptomatology may not be fully excluded, and information on pre-pregnancy depressive symptoms in our study was limited to clinical diagnoses recorded in the medical record. However, we expect minimal influence on our findings since the sensitivity analyses that directly excluded those mothers with previous depression history showed consistent results with our main findings.

Conclusion

We found novel evidence that early pregnancy exposure to higher levels of ambient NO₂, PM_{2.5}, and PM₁₀ was associated with increased risk of moderate-to-high depressive symptoms throughout pregnancy. This finding suggests that air pollution may interfere with early pregnancy adaptations in maternal physiology, including hormonal changes and neuroendocrine processes, potentially altering stress response and mood regulation pathways. We also observed that greater proportions of Black mothers and mothers not cohabitating with a partner during pregnancy were in the moderate-to-high depressive symptom trajectory class, suggesting that race/ethnicity and social support may be important risk factors for prenatal depression trajectories. These findings underscore the importance of addressing both environmental and social determinants to support maternal mental health during pregnancy.

Abbreviations

NO ₂	Nitrogen dioxide
PM _{2.5}	Particulate matter with aerodynamic diameter less than 2.5 µm
HPA	Hypothalamic-pituitary-adrenal
LTCM	Latent class trajectory modeling
hCG	Human chorionic gonadotropin
LGMM	Latent growth mixture modeling
MADRES	Maternal and Developmental Risks from Environmental and Social Stressors
PM ₁₀	Particulate matter with aerodynamic diameter less than 10 µm
O ₃	Ozone
CES-D	Center for Epidemiologic Studies-Depression
DAG	Directed acyclic graph
EMR	Electric medical record
LMR-LRT	Lo-Mendell-Rubin-likelihood ratio test
BIC	Bayesian information criterion
AvePP	Average posterior class membership probabilities
E _k	Relative entropy
RRR	Relative risk ratio
IQR	Interquartile range
NVP	Nausea and vomiting in pregnancy
PUQE-24	24-Hour Pregnancy-Unique Quantification of Emesis and Nausea
GnRH	Gonadotropin-releasing hormone

Supplementary Information

The online version contains supplementary material available at <https://doi.org/10.1186/s12940-026-01314-7>.

Supplementary Material 1.

Acknowledgements

We express our gratitude to the participant families, nurses, midwives, physicians, and staff at our study sites who have cooperated with and participated in the MADRES study. In particular, we extend our appreciation to the dedicated members of the MADRES study team for their relentless efforts to enhance the well-being of our study communities.

Authors' contributions

Conceptualization: YH, TMB; Data curation: TMB, RH, TY, XC; Formal analysis: YH; Funding acquisition: TMB, CB, RH, SFF; Supervision: TMB; Writing- original draft: YH; Writing- review & editing: YH, TY, SPE, XC, ZN, CTC, NP, FL, BG, LAM, EG, SM, JJ, GFD, RH, SFF, CB, TMB. All authors read and approved the final manuscript.

Funding

Maternal and Developmental Risks from Environmental and Social Stressors (MADRES) Center [grant numbers P50ES026086, 83615801–0, P50MD015705] funded by the National Institute of Environmental Health Sciences, the National Institute for Minority Health and Health Disparities, and the Environmental Protection Agency; the Southern California Environmental Health Sciences Center [grant number P30ES007048] funded by the National Institute of Environmental Health Sciences, and the Life course Approach to Developmental Repercussions of Environmental Agents on Metabolic and Respiratory health (LA DREAMERS) [grant number UH3OD023287] funded by the National Institutes of Health Office of the Director ECHO Program.

Data availability

The datasets used and/or analyzed during the current study are available from the corresponding author on reasonable request and after approval by the USC IRB.

Declarations

Ethics approval and consent to participate

The study protocol was approved by the Institutional Review Board (IRB) of the University of Southern California and conducted in accordance with the Declaration of Helsinki and the Belmont Report principles for ethical research involving human subjects.

Consent for publication

Not applicable.

Competing interests

The authors declare no competing interests.

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Received: 24 December 2025 / Accepted: 25 May 2026

Published online: 02 June 2026

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